

JEE MAIN REVISION MASTER: P-BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Boron & Carbon Families (Groups 13 & 14)	17 PYQs	p-Block Elements
High	Nitrogen, Oxygen & Halogen Families (Groups 15, 16, 17 & 18)	8 PYQs	p-Block Chemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Boron & Carbon Families (Groups 13 & 14)	$\text{Borax Bead: } \text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} \xrightarrow{\Delta} 2\text{NaBO}_2 + \text{B}_2\text{O}_3 \text{ (Glassy Bead)}$
Nitrogen, Oxygen & Halogen Families (Groups 15, 16, 17 & 18)	$\begin{aligned} \text{Basicity of } \text{H}_3\text{PO}_2 &= 1, \quad \text{H}_3\text{PO}_3 = 2, \quad \text{H}_3\text{PO}_4 = 3, \\ \text{XeF}_4 &\text{ (Square Planar, } sp^3d^2) \end{aligned}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Boron & Carbon Families (Groups 13 & 14) (Very High Yield — 17 PYQs)

Core Principle: Group 13: Diborane B_2H_6 has two 3-center-2-electron (3c-2e) banana bridge bonds. Borax bead test: $\text{NaBO}_2 + \text{B}_2\text{O}_3$ forms colored metaborates with transition metal salts. Inert pair effect makes +1 oxidation state more stable for Thallium (extTl^+) than +3. Group 14: Diamond (sp^3 , hard insulator) vs Graphite (sp^2 , planar sheets, conductor). Silicones are organosilicon polymers with $[-\text{R}_2\text{SiO}-]_n$ repeating units.

■ **Caution / JEE Pitfall:** Orthoboric acid H_3BO_3 is NOT a protonic acid; it is a weak monobasic Lewis acid that accepts OH^- from water: $\text{B(OH)}_3 + 2\text{H}_2\text{O} \rightleftharpoons [\text{B(OH)}_4]^- + \text{H}_3\text{O}^+$.

■ **Speed Trick:** Due to the inert pair effect, +1 is the most stable state for Tl, and +2 is the most stable state for Pb.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following halides cannot be hydrolysed?..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A metalloid is..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Which of the following compound does not give dinitrogen on heating?..." [Ans: N/A]

3.2 Nitrogen, Oxygen & Halogen Families (Groups 15, 16, 17 & 18) (High Yield — 8 PYQs)

Core Principle: Group 15: Basicity of phosphorus oxoacids: H_3PO_2 is monobasic (has 2 P-H bonds implies strong reducing agent); H_3PO_3 is dibasic; H_3PO_4 is tribasic. Group 16: Contact process for H_2SO_4 with V_2O_5 catalyst. Group 17: Interhalogens XX'_n are more reactive than pure halogens (except F_2). Group 18: XeF_2 (linear), XeF_4 (square planar), XeF_6 (distorted octahedral).

■ **Caution / JEE Pitfall:** Reducing power of H_3PO_2 is highest because it contains TWO direct P-H bonds, whereas H_3PO_4 has zero P-H bonds.

■ **Speed Trick:** Acidic strength of oxoacids of chlorine increases with increasing oxidation state of Cl: $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): " NO_2 gas is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Which one of the following oxoacids of phosphorus can reduce AgNO_3 to metallic silver?..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): " $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating liberates a gas. The same gas will be obtained by..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

NO_2 gas is

- (A) colourless, neutral
- (B) colourless, acidic
- (C) brown, acidic
- (D) brown neutral

Correct Answer: None

Solution Analysis: NO_2 gas is brown and acidic, so the correct choice is: Option C : brown, acidic NO_2 stands for nitrogen dioxide, which is a reddish-brown gas known for its characteristic pungent smell. It is one of the primary pollutants resulting from the combustion of fossil fuels and is a contributor to atmospheric reactions that produce smog and acid rain. As an acidic gas, NO_2 can dissolve in water to form nitric and nitrous acids, contributing to environmental issues such as acid rain.

Benchmark Problem 2: KCET 2024 (Q#3)

A metalloid is

- (A) Bi
- (B) Sb
- (C) P
- (D) Se

Correct Answer: None

Solution Analysis: A metalloid can be defined as chemical elements whose physical and chemical properties lies in between metal and non-metal. Among the given option only antimony (Sb) is metalloid.